Anal 1.

2025.09.10.

a	b	$a \leftrightarrow b$	$\overbrace{a \wedge b}^{A}$	$\overbrace{(\neg a) \wedge \neg b}^{B}$	$A \vee B$
i	i	i	i	h	i
i	h	h	h	h	h
h	i	h	h	h	h
h	h	i	h	i	i

a	b	$a \to b$	$\neg a \vee b$
i	i	i	i
i	h	h	h
h	i	i	i
h	h	i	i

$$\begin{cases} (a \wedge b \wedge c) \to d = \neg(a \wedge b \wedge c) \vee d = \neg a \vee \neg b \vee \neg c \vee d \\ a \to (b \to (c \to a) = \neg a \vee (\neg b \vee (\neg c \vee d)) \end{cases}$$

$$(a \vee b) \wedge c = (a \wedge c) \vee (b \wedge c)$$

$$\neg((a \wedge b) \vee c) \to a) \leftrightarrow b = *$$

$$(a \vee b \vee \neg c) \wedge (a \vee \neg b \vee c) \wedge (\neg a \wedge \neg b)$$

$$\cancel{* = \neg(\neg((a \wedge b) \vee c) \vee a) \leftrightarrow b = ((a \wedge b) \vee c \wedge \neg a \wedge b) \vee}$$

$$\cancel{\vee ((\neg(a \wedge b) \wedge \neg c \vee a) \wedge \neg b) = ((a \wedge b) \vee c \wedge \neg a \wedge b) \vee}$$

$$\cancel{\vee ((\neg a \vee \neg b) \wedge \neg c \vee a) \wedge \neg b) = ((a \vee c) \vee (b \wedge c) \wedge \neg a \wedge b) \vee}$$

$$\cancel{\vee ((\neg a \wedge \neg c) \vee (\neg b \wedge \neg c) \vee a}$$

a	b	c	A	B
i	i	i	h	
h	i	i	i	
i	h	i	i	
h	h	i	h	
i	i	h	h	
h	i	h	i	
i	h	h	i	
h	h	h		

Minden medve szereti a mézet.

M mézek hza B medvék hz-a

S szeret ~~amit~~

$\exists m \in M, \exists b \in B, \neg S(b,m)$

Minden asszony életében $\exists$ pillanat, amikor olyat akar tenni, amit nem szabad

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$\varphi(b,k)$ boha megcsípte a kecskét

$\forall k \in K, \exists b \in B, \varphi(b,k)$

tag. $\exists k \in K, \forall b \in B, \neg \varphi(b,k)$

$\exists k \in K, \forall b \in B, \varphi(b,k) : \forall k \in K, \exists b \in B, \neg \varphi(b,k)$

$\forall b \in B, \exists k \in K, \varphi(b,k) : \exists b \in B, \forall k \in K, \neg \varphi(b,k)$

$$(A\wedge B)\to(C\vee D)=(\neg A\vee\neg B)\vee C\vee D$$

$$(A\oplus B)\vee(C\wedge D)=\neg\big((A\wedge B)\vee(\neg A\wedge\neg B)\big)\vee(C\wedge D)$$

$$\neg\big(\underbrace{(P\wedge Q)\vee(\neg P\wedge\neg Q)}\big)\wedge\big(R\vee\neg\underbrace{(S\wedge T)\vee(\neg S\wedge\neg T)}\big) \quad \boxed{\leftrightarrow}$$

$$(P\wedge\neg Q)\vee(\neg P\wedge Q)\wedge\big(R\vee(S\wedge\neg T)\vee(\neg S\wedge T)\big) \quad \boxed{\oplus}$$

$$\neg(P\Leftrightarrow Q)=P\oplus Q$$

2025.09.17.

gratulálok szereztél egy tevét  +1 pont

$$\left(\bigcup_{i\in I}A_i\right)\cap\left(\bigcup_{j\in J}B_j\right)=\bigcup_{i\in I}\bigcup_{j\in J}(A_i\cap B_j)$$

$$\exists i\in I\ \exists j\in J\ \ e\in A_i\wedge e\in B_j \qquad \exists i\in I\ \ e\in\bigcup_{j\in J}(A_i\cap B_j)\to$$

$$\to\exists i\in I\ \exists j\in J:\ e\in A_i\cap B_j$$

C egy alaphalmaz

$$C-\bigcup_{i\in I}A_i=\bigcap_{i\in I}(C-A_i)$$

$$e\in C\wedge\forall i\in I: e\notin A_i \qquad \forall i\in I: e\in C\wedge e\notin A_i$$

$x\in\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty}A_k \iff x$ véges sok kivétellel $\forall A_k$-ban benne van

$x\in\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty}A_k \iff x$ végtelen sok A_k-ban benne van

$\boxed{\rightarrow}$ kontrapozíció

$\boxed{\leftarrow}$

$$\exists i\in\{1,\dots,n\}\ \forall j\in\{k,\dots,n\}: x\in A_k$$

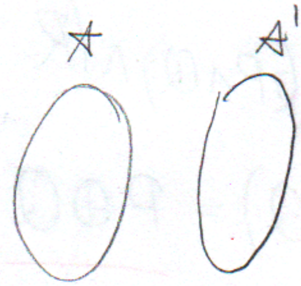
$$\forall k\in\{1,2,\dots\},\ \exists j\in\{k,k+1,\dots\}: x\in A_j$$

X hz $H=\{0,1\}$ $p: H\times H \to H$ $p(a,b)=p(b,a)$ $A\subset X$

$A' = \{x\in X \mid \forall a\in A: p(x,a)=1\}$

a) $\forall A,B\subset X$ $A\subset B \to B'\subset A'$

b) $\forall A\subset X$ $A\subset A''$



c) $A'=A'''$

d) $H=\mathbb{R}$

e) $H=\{1\}$

---

e) $X'=X$

a) $(x\in B \to x\in A) \to$

b) A' azon elemek, melyekhez p 1-et rendel.

$p(a',a)=1$

$p(x,a'')=1$

---

a1) $A = \bigcup(\mathcal{P}(A))$

a2) $\bigcup B\subset A \leftrightarrow B\subset \mathcal{P}(A) \leftrightarrow B\in\mathcal{P}(\mathcal{P}(A))$

b) $\{x,y\}\in A \to x,y\in \bigcup A$

d) f fv $Ran(f), Dom(f)\in \mathcal{P}(\bigcup\bigcup f)$

c) $(x,y)\in A \to x,y\in\bigcup\bigcup A$

~~d) f fv. Ran(f), Dom(f)~~

e) $f: A\to B$

$f\in\mathcal{P}(\mathcal{P}(\mathcal{P}(A\cup B)))$

1) A hz $\to \{A\}$

2) A, B hz $\to \{A, B\}$

3) A, B, C hz $\to \{A, B, C\}$

---

c) $(A \wedge \neg B) \vee (\neg A \wedge \neg C) \vee (B \wedge C) \vee (A \oplus C)$

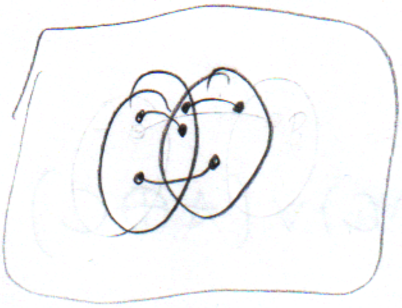
A	B	C	$A \wedge \neg B$	$\neg A \wedge \neg C$	$B \wedge C$	$A \oplus C$	
1	1	1	0	1	1	1	1
1	1	0	0	0	1	0	1
1	0	1	1	1	0	1	1
1	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	0	0	0	0	1	1
0	0	1	0	0	0	0	0
0	0	0	0	0	0	1	1

$(C \vee A)$	$(C \vee B)$
1	1
1	1
1	1
1	0
1	1
1	0
1	1
0	0
0	

$A, B \subset X$

2025.09.24.

$R[A] \in B \Longleftrightarrow R^{-1}[X-B] \subset X-A$



$R^{-1}[X-B] \not\subset X-A \longrightarrow$

$\exists x \in R^{-1}[X-B] \wedge x \notin X-A \Longleftrightarrow$

$\longrightarrow$

$\exists a \in R^{-1}[X]$

$\mathcal{H} \subset \mathcal{F}(A,B)$ $f: A \xrightarrow{?} B$

$h_1, h_2 \in \mathcal{H}$, $h_1 \subset h_2 \vee h_2 \subset h_1$

$\bigcup \mathcal{H} \in \mathcal{H}$

$\exists i \in I,\ h_i \in \mathcal{H},\ x \in h_i$

a) $f = \bigcup \mathcal{H}$ függvény

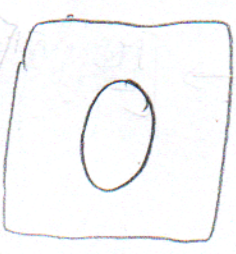
$(a,c),(a,b) \in f \rightarrow \exists h_i, h_j \in \mathcal{H}: (a,c) \in h_i, (a,b) \in h_j \rightarrow$

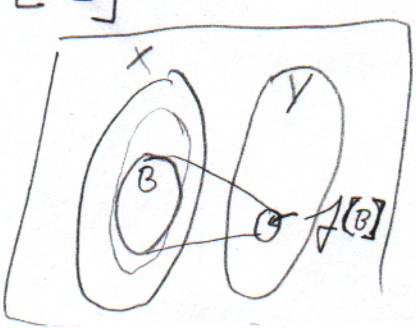
$(h_i \subset h_j \rightarrow b \neq c) \vee (h_j \subset h_i \rightarrow b \neq c)$

b) részről $(b,a),(c,a)$-ra ugyanez

$f: X \xrightarrow{?} Y$ $\quad A \subset X \rightarrow A \subset f^{-1}[f[A]]$

$B \subset Y \rightarrow f[f^{-1}[B]] \subset B$





f szürjektív, akkor

$x_n \trianglelefteq y_n \Leftrightarrow (x_n - y_n \to 0) \vee (x_n < y_n \ \forall n$-re véges sok kivétellel$)$

i) $\trianglelefteq$ reflexív

ii) nem

iii) nem

iiii) nem

$\mathbb{Q}$

$\mathbb{R}$

$\mathbb{Z}$

nem Cauchy

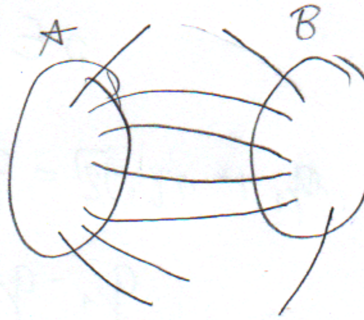
$(-2)^n \quad (-2)^{n+1}$

v) $(-1)^n + \frac{1}{n} \trianglelefteq (-1)^n + \frac{1}{n^2}$

$1 + \frac{2}{n^2}$

2025.10.01.

$R[A] \subset B \Leftrightarrow R^{-1}[B^c] \subset A^c$



4.1. $]0,1] \to [0,1]$

$f: \overbrace{x \mapsto x}^{\text{ez id.}}$ ha $x \in [0,1] - \mathbb{Q}$

$r_0, r_1, r_2, \dots$ a racionálisak

$\searrow r_1, \searrow r_2, \searrow r_3$

tehát $r_k \mapsto r_{k+1}$ tehát „arrébtoljuk"

d) $[2,5] \to]10,20[$

id helyett $\frac{10}{3}x + \frac{10}{3}$, felsoroljuk (elhagyjuk) a $\mathbb{Q}$-kat és a 10-et és 20-at ha $\mathbb{Q}$-beli: $r_k \longmapsto \frac{10}{3} r_{k+1} + \frac{10}{3}$

vagy $]-1,1[\longrightarrow [-1,1]$

$$f(x) = \begin{cases} -x - \frac{3}{2} & x \in]0, -\frac{1}{2}] \\ -x + \frac{3}{2} & x \in [\frac{1}{2}, 1[\\ 0 & \end{cases}$$

valami Cantor halmazosan (nemtudom elolvasni)

$$R(f) = [-1, -\frac{1}{2}[\cup \{0\} \cup]-\frac{1}{2}, 1]$$

e) $f: \mathbb{R} - \mathbb{Q} \xrightarrow{b} \mathbb{R}$

$\mathbb{Q} \subset H = \{ \underbrace{q}_{\in \mathbb{Q}} + n\sqrt{2} \quad n \in \mathbb{N}_0 \}$

$\mathbb{R}$-ből $(\mathbb{R} - \mathbb{Q})$-ba volt ~~ez~~

$\mathbb{R} = H \cup \mathbb{R} - H \qquad f|_{\mathbb{R}-H} = id$

bijekció-e

$\in H \quad q + n\sqrt{2} \longmapsto q + (n+1)\sqrt{2}$

$$q_1 + n\sqrt{2} = q_2 + m\sqrt{2}$$

$$q_1 - q_2 = (m-n)\sqrt{2}$$

4.2. trivi $\frac{1}{2}$ oldalú négyzetek

4.3. $1, a_1, a_2, a_3, \ldots$

megoldottam, majd leírom, mert táblánál írtam

(b) $\left|\bigcup_{x\in\mathbb{R}} A_x\right| = |\mathbb{C}|$ de adjunk meg bijekciót

$\operatorname{arctg}(x) \xrightarrow{\alpha} \measuredangle$

$\mathbb{R} \longmapsto$ szakasz $\longmapsto$ szögre

(4.7) a) $2\mathbb{N}$

b) $2\mathbb{N}+1$

c) $\{2^k \mid k\in\mathbb{N}\}$

d) $\{n\in\mathbb{N} \mid n \text{ prím}\}$

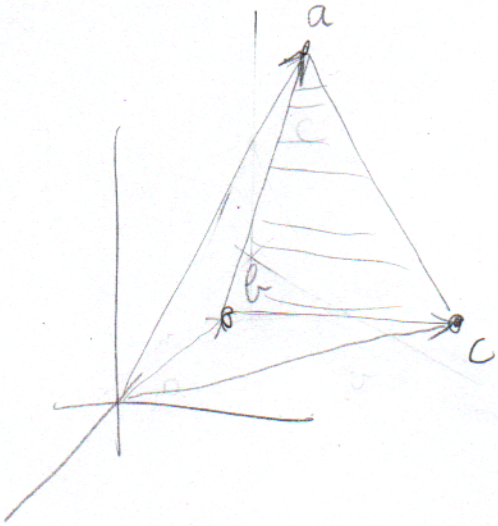
lajháról nem tudok rajzolni

nincs pluszpont.

(4.8) $(a_n, \dots, a_0)$ egész együtthatókból álló véges sorozat

(4.9.) trivi

(4.10.)



π-hez nincs, sőt $\mathbb{R}-\mathbb{Q}$-hoz sem, mivel:

$T = \langle \underbrace{a-b}_{\in\mathbb{Q}}, \underbrace{c-b}_{\in\mathbb{Q}} \rangle \in \mathbb{Q}$

(4.11.) a) $\underbrace{\aleph_0 + \mathfrak{c}}_{\text{az összes}}$ b) $\underbrace{\mathfrak{c}+\mathfrak{c}}_{\mathfrak{c}}$ c) $\underbrace{\aleph_0 \mathfrak{c}}_{\mathfrak{c},\ \aleph_0}$ d) $\underbrace{\mathfrak{c}\mathfrak{c}}_{\mathfrak{c}}$

2025.10.08.

5.1 $\quad x^n \quad [0,1]$

határfüggvény

$\forall b<1$-re $[0,b]$ $\quad x^n \rightrightarrows x$

$|f_n - f| < b^n$

$|x^n - 0| < b^n \rightrightarrows 0$

univerzális küszöb

ez a határfv

5.2 a) $(\log x)^n \quad \left(\frac{1}{e}, e\right]$

határfv

b) $n \sin\frac{x}{n} = x\dfrac{\sin\frac{x}{n}}{\frac{x}{n}} \rightrightarrows x \quad \mathbb{R}$

c) $\dfrac{x^n}{1+x^{2n}} \quad |x|<1 \to \dfrac{x^n}{1+x^{2n}} \rightrightarrows 0$

$\mathbb{R}-\{-1\}$

d) $\dfrac{2nx^4}{n^2x^4+n+3} = \dfrac{\frac{2}{n}x^4}{x^4+\frac{1}{n}+\frac{3}{n^2}} \quad \mathbb{R}$

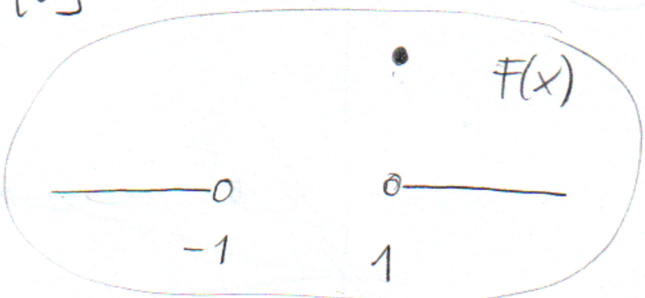
$\frac{2}{n} \to 0, \quad \frac{1}{n} \to 0, \quad \frac{3}{n^2} \to 0, \quad n^2x^4+n+3 \to 0$

5.3 a) $f_n(x) = \dfrac{1}{x^n} \quad Dom(f_n) = \mathbb{R}-\{0\}$

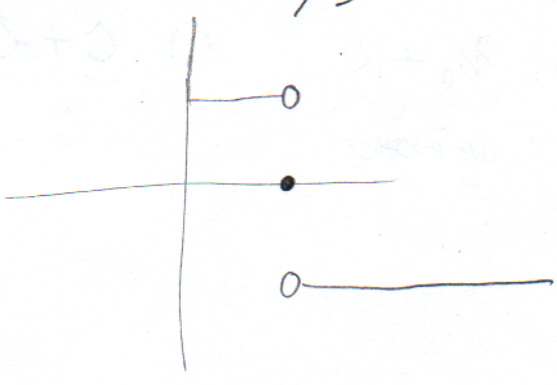
$\left|\dfrac{1}{x}\right| < 1 \to \dfrac{1}{x^n} \rightrightarrows 0$

$\left|\dfrac{1}{x}\right| > 1$

$x = 1 \to \dfrac{1}{x^n} \rightrightarrows 1$

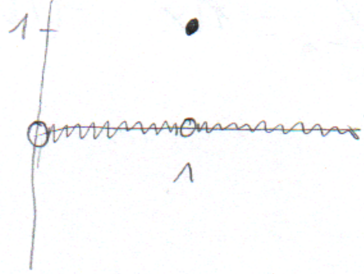


$\not\rightrightarrows$



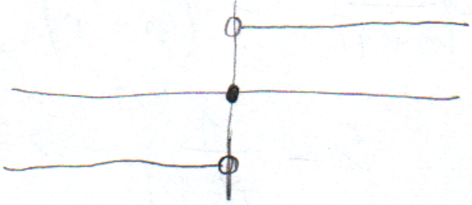
b) $\dfrac{1-x^{2n}}{1+x^{2n}}$ páros

c) $\frac{2x^{2n}}{1+x^{4n}}$ páros $|x|<1$



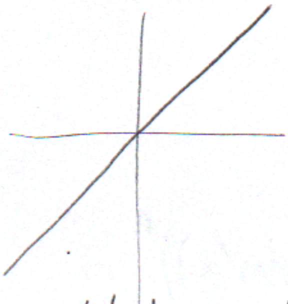
$\not\rightrightarrows$

d) $\frac{e^{nx}-1}{e^{nx}+1}$



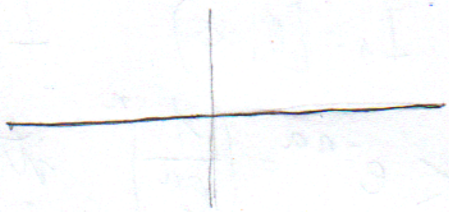
$\not\rightrightarrows$

e) $n\log\left(1+\frac{x}{n}\right)=x$ $\mathcal{H}_H=\mathbb{R}$



$\not\rightrightarrows$ mi a pipáért nem?

f) $\frac{1}{(x+n)^2}\rightrightarrows 0$

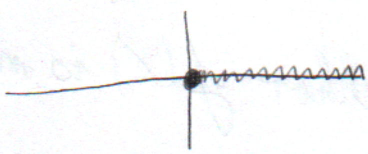


$\not\rightrightarrows$

g) $\sum_{k=1}^{n}\frac{1}{k^x}$ $x>1$ / $a>1$ $[a,\infty)$

itt egyenletesen monoton egyébként nem

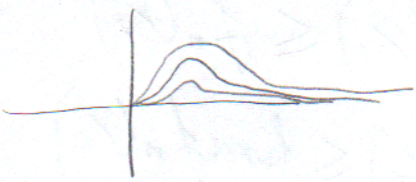
l) xe^{-nx}



$f_n'=e^{-nx}+x(-n)e^{-nx}=$

$=e^{-nx}(1-nx)$

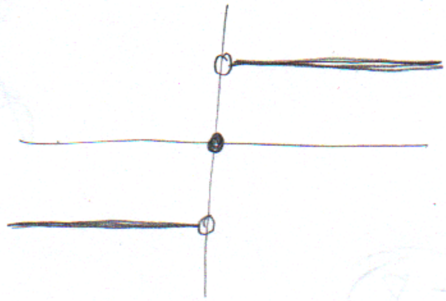
mi?



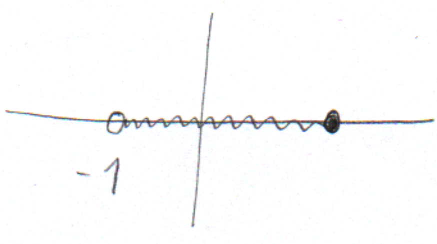
$f_n\left(\frac{1}{n}\right)=\frac{1}{n}e^{-1}$

i) $\frac{2}{\pi}\operatorname{arctg}(nx)$ $x>0, f_n\rightrightarrows 1$

$x=0$



j) $x^n-x^{n+1}=x(x^n-1)$



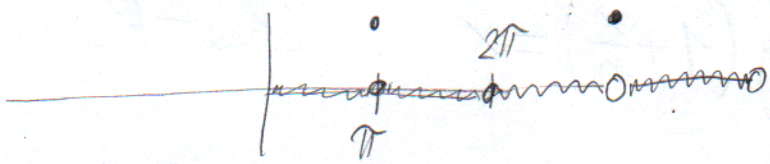
$(-1,1]$

$$f_n' = nx^{n-1} - (n+1)x^n = x^{n-1}(n-(n+1)x)$$

$$\frac{n}{n+1} = x$$

$$\left(\frac{n}{n+1}\right)^n - \left(\frac{n}{n+1}\right)^{n+1} = \left(\frac{n}{n+1}\right)^n \left(1 - 1 + \frac{1}{n+1}\right)$$

$$\left(1 - \frac{1}{n+1}\right)^n \to \frac{1}{e} \qquad |f_n - 0| < \frac{1}{n+1}$$



k) $(\sin(x))^n$

5.4 a) e^{-nx} $\quad I_1 = [0,\infty) \quad I_2 = [a,\infty)$

$$|e^{-nx} - 0| < e^{-na} = \left(\frac{1}{e^a}\right)^n$$ itt venni kell vmi küszöbindexet

5.7 $f_n : I \subset \mathbb{R}$ mon. növ. függvényekből álló

$\lim_{n\in\mathbb{N}} f_n(x) = f(x)$, akkor $f(x)$ is monoton növekvő!!

$$f_n(x_1) \le f_n(x_2) \quad \text{ha } x_1 \lessgtr x_2$$

$$\lim f_n(x_1) \le \lim f_n(x_2)$$ ~~azaz~~ a lim monoton

$$f(x_1) \le f(x_2)$$

5.8

$\longrightarrow$

$g:[0,1] \subset\!\!\to \mathbb{R}$ folyt. fv.

$f_n:[0,1] \subset\!\!\to \mathbb{R} \quad f_n(x)=g(x)x^n \quad (n\in\mathbb{N}^+)$

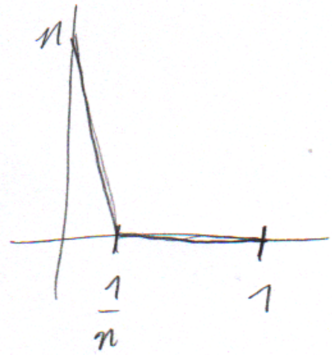
$g(1)\neq 0 \to f_n(1) \to\!\!\!\to g(1)\neq 0$ tehát van szakadási pont,

tehát ~~nem~~ $\not\rightrightarrows$

$g(1)=0 \longrightarrow \quad |g(1)| < \varepsilon$

még nem értem
csak annyit, hogy
ε-hoz kell küszöbindexet
keresni és igenis
számít, hogy g felveszi
a maximumát.

5.9. $\lim\limits_{n\to\infty} \int_2^5 nx\, e^{-nx} = \int_2^5 \lim\limits_{n\to\infty} nxe^{-nx} = 0$



$$\left| x\frac{n}{(e^x)^n} - 0 \right| \leq \frac{5n}{2^{2n}}$$

$$\lim \int_0^{2\pi} \frac{\operatorname{arctg}(n^5x^2)}{x+\sqrt{n}} = 0$$

$$\left| \frac{\operatorname{arctg}(\ )}{x+\sqrt{n}} \right| < \frac{\frac{\pi}{2}}{\sqrt{n}}$$

$$\lim \int_0^1 \frac{x^n}{e^{nx}} = 0$$

$$\left| \frac{x^n}{e^{nx}} - 0 \right| < \frac{1}{2^n} \quad x<\frac{1}{2}$$

$$< \frac{1}{\sqrt{e^n}} \quad x \geq \frac{1}{2}$$

$\max(N_1, N_2)$

$$\lim \int_0^1 \frac{x^n}{1+x}\,dx$$

$\frac{1}{2}$

$g(x)=\frac{1}{1+x}$